

## Scalability & Future Plan

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|---------------|--------------------------------------------------------------|
| Date          | 02 July 2024                                                 |
| Team ID       | SWTID-2026-4836                                              |
| Project Name  | OptiCrop: Smart Agricultural Production Optimization Engine. |
| Maximum Marks | 1 Mark                                                       |

### Scalability & Future Plan:

The current **OptiCrop** system is designed as a web-based crop recommendation platform that can be easily scaled to support a larger number of users and additional functionalities in the future.

### Current System Limitations:

| S.No | Limitation                                                   | Impact                                                      | Priority to Address (High / Medium / Low) |
|------|--------------------------------------------------------------|-------------------------------------------------------------|-------------------------------------------|
| 1    | Supports only a limited agricultural dataset.                | May reduce prediction accuracy for other regions and crops. | High                                      |
| 2    | Application is currently designed for local deployment only. | Cannot support a large number of simultaneous users.        | Medium                                    |
| 3    | No user authentication and cloud database integration.       | Limited security and scalability.                           | Medium                                    |

### Scalability Plan:

| S.No | Scalability Aspect | Current State                                           | Proposed Upgrade / Solution                                                   |
|------|--------------------|---------------------------------------------------------|-------------------------------------------------------------------------------|
| 1    | User Load          | Supports a small number of users on a local server.     | Deploy on cloud platforms such as Render, AWS, or Azure with load balancing.  |
| 2    | Data Storage       | Uses local CSV dataset and model files.                 | Integrate MySQL or MongoDB for large-scale agricultural data storage.         |
| 3    | Performance        | Predictions are generated using a locally stored model. | Optimize model performance and use caching techniques for faster predictions. |
| 4    | Security           | Basic input validation is implemented.                  | Add user authentication, HTTPS, and secure API integration.                   |

Future Roadmap:

| Phase   | Planned Feature / Enhancement               | Target Timeline | Expected Impact                                                              |
|---------|---------------------------------------------|-----------------|------------------------------------------------------------------------------|
| Phase 1 | Cloud deployment and online hosting         | 2 Months        | Improve accessibility and support more users.                                |
| Phase 2 | Real-time weather API integration           | 3 Months        | Provide more accurate crop recommendations.                                  |
| Phase 3 | Mobile application development              | 4 Months        | Enable farmers to access the system through smartphones.                     |
| Phase 4 | Advanced AI models and multilingual support | 6 Months        | Increase prediction accuracy and usability for farmers in different regions. |